



Polystyrene/ ZnO Nanocomposite Films with Optimized Optical Properties for UV-shielding Applications

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Abstract

Nanostructured dispersed polymeric materials were used a lot in various fields of application due to their improved structural, mechanical, optical and electrical properties than the base polymeric materials. In present research work, ZnO doped polystyrene (PS) films were synthesized at room temperature via solution-cast technique. The characterizations of these nanocomposite films were done using Ultraviolet Visible spectroscopy (UV–Visible), RAMAN, Photoluminescence and X-Ray Diffraction (XRD) techniques. By characterization, the change in optical and structural properties of PS nanocomposite films at different concentrations can be observed. Different parameters like theoretical & experimental density, direct band gap energy, absorption coefficient, generation of trap centres, change in crystallinity along with various structural parameters are reported in this work. This will help in applications of these materials in UV-shielding and optoelectronic devices.

Keywords Nanocomposite films · Polystyrene · Zinc oxide nano powder · Optical properties · Photoluminescence · X-ray diffraction · UV-shielding

1 Introduction

Organic–inorganic nanocomposite materials attained the attention of different research groups in last few decades [1–3]. These composites have different mechanical, optical, chemical and physical properties as compared to their parent polymer and nano filler. Polymers are suitable host matrix for developing nanocomposite materials because of their simple synthesis processes, low cost, light weight, moisture and corrosion resistant [4, 5]. These nanocomposites have advanced applications in different fields like solar cells, UV-shielding materials, sensors, photocatalysis, hybrid OLEDs and optoelectronics etc. [6–9]. Polystyrene (PS) draws our attention among various polymers due to its unique properties like light weight, cost-effectiveness, transparency and optical stability. It has excellent biocompatibility, chemical, electrical and mechanical stability. It is stiff, easily processable, recyclable and reusable material [10, 11]. The structure of PS polymer contains aromatic ring along with an attached aliphatic chain. This aromatic ring has delocalized electrons

and these electrons can be manipulated to make this polymer conducting. It can also work as a perfect UV- absorber owing to its advanced optical properties [12–14].

Among variety of inorganic nanofillers, ZnO is highly crystalline, direct wide band gap and environmental friendly semiconductor material. It is a non- toxic material which possess the large excitation binding energy (= 60 meV) and band gap of 3.37 eV [15, 16]. ZnO is pyroelectric & piezoelectric material useful in transparent conductors, gas sensors, dye-sensitized devices etc. [17–20]. The different nanostructures of ZnO are responsible for modifications in structural, morphological, electrical and optical properties of nanocomposites. As ZnO has large surface to volume ratio it give a better interfacial coordination between polymer and nanoparticles leading to enhanced optical properties and crystallization degree [1, 21]. Introduction of ZnO nanoparticles into polymer matrix make them interesting towards technological optoelectronic applications due to the unique properties of ZnO.

Development of thin organic–inorganic nanocomposite films for UV- shielding applications has been enduring the interest of various researchers from a long time. UV-radiations especially UVB of range 280–320 nm are more harmful for both living beings and organic substances than UVA radiations (320–400 nm) due to their high energy. In living

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beings, these radiations damage DNA cells and causes skin cancer. In organic substances like polymers, dyes, semiconductors etc. UVB radiations causes degradation [21–23].

In the present work, ZnO doped PS nanocomposite films were prepared via solution-mixing technique. PS/ZnO nanocomposite films with 0, 5, 10, 15 and 20 wt% of ZnO are abbreviated as PS-1, PS-2, PS-3, PS-4 and PS-5 respectively. The significant changes in optical and structural properties of these nanocomposite films were observed at room temperature by varying ZnO concentration into PS matrix. Here our main emphasis is to develop a thin, self-supporting, flexible and transparent UV-shielding films.

2 Experimental

2.1 Materials

Zinc Acetate dihydrate (99.9%, Merck Specialities Pvt. Ltd., Mumbai, India), Polystyrene powder (Polysciences Pvt. Ltd.), Sodium Hydroxide, Ethanol, Distilled water and Toluene were used for the synthesis of PS/ZnO nanocomposite films. All these purchased chemicals are of A.R. grade with 99.9% purity.

2.2 Synthesis of PS/ZnO Nanocomposite Films

For the preparation of pristine PS and PS/ZnO nanocomposite films, 2 gm polystyrene powder was added to 50 ml toluene in a beaker. The solution was stirred for 30 min using magnetic stirrer at 600 rpm and sonicated for 1 h using Probe Sonicator to get the homogeneous solution. The solution was poured to a plain petri dish and then placed on a balanced dust free chamber at room temperature for 24 h. The pristine PS film was obtained after the complete evaporation of the solvent. Repeat this process by incorporating suitable amount of ZnO (i.e. 5, 10, 15 & 20 wt%) and again sonicate the solution to obtained ZnO dispersed PS solution. After pouring and evaporation of solvent, PS/ZnO nanocomposite films were obtained.

2.3 Characterizations

The optical analysis of prepared pristine and nanocomposite films was done via UV–Visible, RAMAN and Photoluminescence (PL) spectroscopic techniques. The absorption spectrums of prepared films were recorded on Bench-top UV-3600i Plus (Shimadzu) UV–Visible NIR spectrophotometer having both double beam and monochromator. The RAMAN spectrums were recorded on RAMAN spectroscopy (WITEC ALPHA 300) in the range 0 to 3500 cm^{-1} . This system has combination of two lasers with 535 nm & 325 nm wavelengths for observing proper RAMAN

spectrum of the samples. Photoluminescence (PL) spectra of PS/ZnO nanocomposite films were recorded using Luminescence Spectrophotometer LS55 Perkin Elmer USA having a source of Xenon lamp and two different monochromators. The X-Ray diffraction patterns of these nanocomposite films were recorded on XRD-6000 Shimadzu Analytical diffractometer. The X-ray tube contains voltage source of 40 kV with target element Cu. These films were scanned from 10 to 80° with a continuous scan rate 2 deg/min.

3 Results and Discussion

3.1 Density Measurement of Films

The density measurement of prepared pristine PS and PS/ZnO films has been carried out experimentally using Analytical Weighing Balance (Mettler Toledo ME204). These measurements show that the density of PS films (ρ_{exp}) increases with the increase in filler concentration (Table 1). The change in properties of nanocomposites with volume fraction of filler content was reported by Vaishnav et al. [24].

For theoretical calculations of the density, the volume fraction of nanofillers can be obtained by using equation

$$V_f = \frac{W_f}{W_f + W_p \left(\frac{\rho_f}{\rho_p} \right)}$$

where V_f & W_f are the volume and weight fraction of nanofillers, W_p is the weight fraction of polymer matrix, ρ_f & ρ_p are the densities of nanofillers and polymer matrix respectively.

For present calculations, the densities of Polystyrene polymer and ZnO nanofillers have been considered as 1.02 g/cc and 5.61 g/cc respectively. Using Rule of mixture, the theoretical density of polymer nanocomposites can be calculated as

$$\rho_{\text{theo}} = \rho_f V_f + \rho_p V_p$$

where V_p is the fraction of polymer matrix.

Table 1 Filler concentration and density of PS/ZnO nanocomposite thin films

Samples	Filler concentration		ρ_{exp} (g/cc)	ρ_{theo} (g/cc)	$\rho_{\text{exp}}/\rho_{\text{theo}}$
	By wt%	By Vol%			
PS-1	0	0	0.99	1.02	0.97
PS-2	5	0.91	1.10	1.06	1.04
PS-3	10	1.90	1.12	1.11	1.01
PS-4	15	2.99	1.18	1.16	1.02
PS-5	20	4.187	1.24	1.21	1.02

Theoretical values calculated for synthesized samples at different wt% or vol% are shown in Table 1 and it confirms that the experimental & theoretical values are in good agreement with each other.

Due to the higher density of ZnO, the density of nanocomposite films was changed from 1.02 g/cc (for pristine PS) to 1.21 g/cc (for 20 wt% ZnO nanocomposite film).

3.2 UV–Visible Spectroscopy

UV–Visible spectroscopy is an eminent technique for investigating optical properties. The UV–Visible spectrums were recorded for PS/ZnO nanocomposite films in absorption mode. It was done to analyze the effect of ZnO doping into PS matrix in terms of optical advancement. In the present investigation, absorption spectra of nanocomposite films was recorded in the wavelength range of 250–600 nm. The UV–Visible spectra of pristine PS and PS/ZnO nanocomposite films has been shown in Fig. 1. The absorption phenomenon is related to the excitation of electrons and is dominated in nanocomposite films. The excitation of electrons is from the valance band to the conduction band and optical energy band gap can be calculated. The small absorption edge is observed at 300 nm due to π - π^* electronic transition in delocalized C=C among phenyl group of PS. Shift in the wavelength of this peak is observed with increase in ZnO concentration. It shows the coordination bonding between carbonyl group of PS and ZnO nano powder. The spectra of PS/ZnO nanocomposite films also shows another absorption peak at 372 nm indicating the presence of ZnO in PS matrix. This peak becomes dominant with the increase in filler concentration. The similar peak position corresponding to ZnO was reported by Samuel M. et al. [25]. No change

was observed in the peaks position of pristine PS & ZnO, after doping into the PS matrix.

The incorporation of ZnO into PS polymer has been modified the ligands of filler and host matrix and increased the UV-shielding efficiency of the composite films. Their proper dispersion will leads the development of optically stable material [21].

To calculate the optical energy band gap Tauc's relation has been used [4, 26],

$$\alpha h\nu = c(h\nu - E_g)^n$$

where α is absorption coefficient, $h\nu$ is energy of photon, E_g is energy band gap, c is constant. The value of n can be $\frac{1}{2}$ and 2 to calculate direct and indirect energy band gap of the materials respectively. The calculation of optical band gap energy has been carried out using absorption coefficient (α) as

$$\alpha = 2.303 \log \frac{A}{t}$$

where A is absorption and t is the thickness of film.

The optical band gap energy is measured conventionally by linear extrapolation of the graph among modified Kubelka–Munk function $(\alpha h\nu)^2$ and energy ($h\nu$). This method gives lower values of E_g and may lead to less accurate results. However, the baseline corrected method provides more accurate results for nanocomposite films. Here the intersection of two slopes results into the value of energy band gap for different samples [27]. The optical band gap of ZnO nano powder and PS/ZnO nanocomposite films as a function of ZnO concentration has been illustrated in Fig. 2. The direct band gap of nanocomposite films estimated via direct application and baseline corrected simplified analysis of Tauc's plot were collectively tabulated in Table 2.

The variation in band gap of PS/ZnO nanocomposite films from 4.45 to 3.09 eV (by direct extrapolation of Tauc's plot) and 4.48 eV to 3.04 eV (Simplified analysis using baseline correction of Tauc's plot) were observed with change in concentration of ZnO among PS matrix which may be due to the complexation or cross-linking among amorphous phase of PS and crystalline phase of ZnO nano powder [28, 29]. From the above results, PS-1 can absorb light of wavelength (< 280 nm) but nanocomposite films i.e. PS-2, PS-3, PS-4 & PS-5 can absorb the lights of wavelength 280–415 nm. It indicates that the synthesized nanocomposite films has the tendency to absorb UVB radiations and are perfect materials for UV-protective coatings and optical fibres [22]. Moreover, the transparency of 5wt% ZnO doped nanocomposite film is very close to pristine PS, therefore more suitable for UV-shielding applications.

The relationship among absorption coefficient (α) and energy ($h\nu$) for different PS/ZnO nanocomposite films were

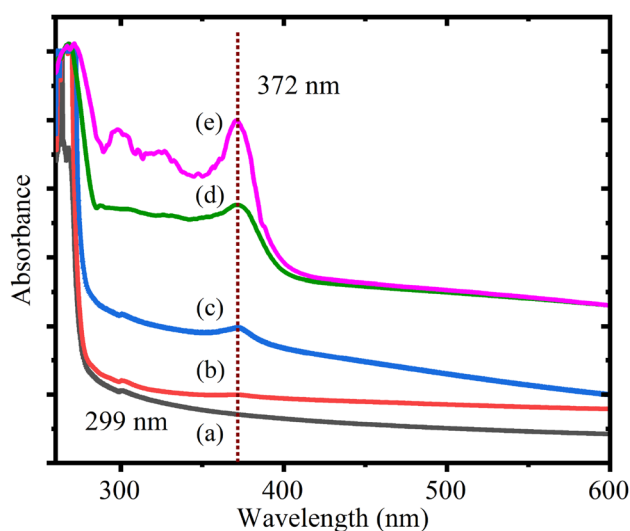


Fig. 1 UV–Visible spectra of **a** PS-1, **b** PS-2, **c** PS-3, **d** PS-4, **e** PS-5 nanocomposite films

Fig. 2 Tauc's plot between Energy ($h\nu$) and $(\alpha h\nu)^2$ for ZnO nano powder and PS/ZnO nano-composite films for estimating optical energy band gap

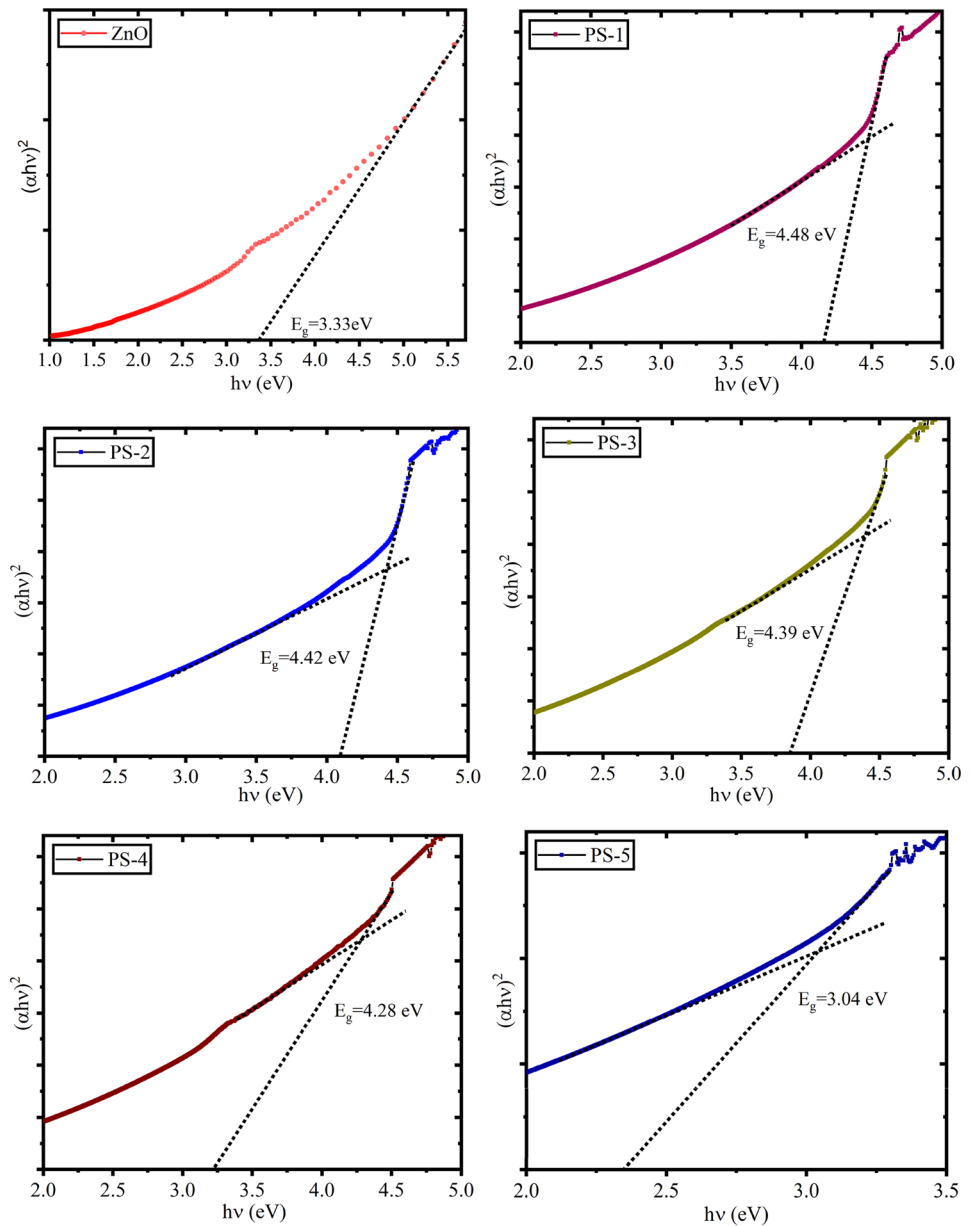


Table 2 Experimentally observed E_g values using conventional and simplified analysis of Tauc's plot

Samples	E_g (Calculated using conventional extrapolation of Tauc's plot) (eV)	E_g (Calculated using baseline correction of Tauc's plot) (eV)
PS-1	4.45	4.48
PS-2	4.36	4.42
PS-3	4.24	4.39
PS-4	3.96	4.28
PS-5	3.01	3.04

depicted in Fig. 3. It shows the variation in absorption coefficient from low energy to high energy. At lower energies i.e. $h\nu < E_g$, electron can't jump from valance band to conduction

band due to insufficient absorption energy. At higher energies i.e. $h\nu > E_g$, the value of absorption coefficient is large. It leads to the higher possibility of electronic transitions

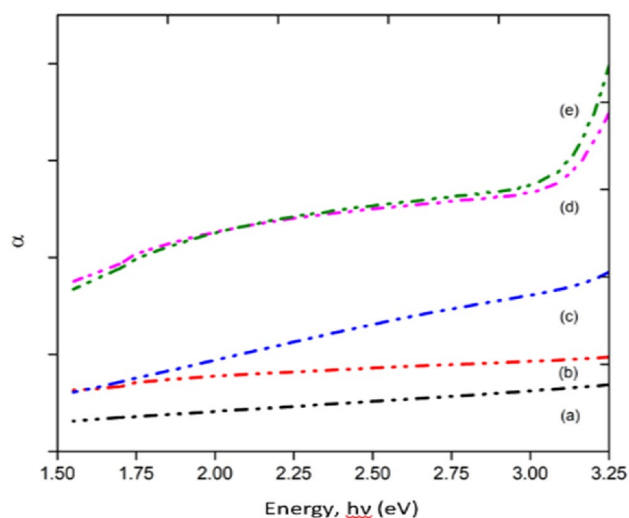


Fig. 3 Change in optical absorption coefficient with energy for **a** PS-1, **b** PS-2, **c** PS-3, **d** PS-4 and **e** PS-5 nanocomposite films

between valance band and conduction band. It results in increase in conductivity with increasing ZnO concentration.

3.3 RAMAN Spectroscopy

Raman spectroscopy is an interesting technique to know about composition, crystallization, molecular disorderness

and vibrations of nanocomposite films. This technique is dedicated to analyze the effect of ZnO nanoparticles on PS polymer by varying its content. Figures 4 and 5 depicts the Raman spectra of pristine and nanocomposite films.

Raman peaks observed at 12 cm^{-1} , 83 cm^{-1} (due to lattice vibrations of complete crystal), 630 cm^{-1} (deformation in benzene ring), 805 cm^{-1} (C-H out-of-plane deformation), 1011 cm^{-1} (aromatic ring breathing), 1040 cm^{-1} (C-H in-plane deformation), 1163 cm^{-1} (C-C stretching vibration), 1205 cm^{-1} , 1325 cm^{-1} (C-H, C=C vibrational modes), 1455 cm^{-1} (-CH₂ chain-scissoring), 1590 cm^{-1} (C=C stretching), 1610 cm^{-1} (ring-skeletal stretching vibration), 2856 cm^{-1} (symmetric & asymmetric stretching of >CH₂ methylene group in aliphatic chain), 2913 cm^{-1} , 2983 cm^{-1} (C-H vibrations in aliphatic chain) and 3054 cm^{-1} , 3065 cm^{-1} (C-H vibrations in aromatic ring) for PS and PS/ZnO nanocomposite films. A small hump was observed corresponding to raman shift 417 cm^{-1} in PS/ZnO nanocomposite films but absent in case of PS shown in inset (Figs. 4 and 5). Similarly, a hump around 2548 cm^{-1} is observed in the pristine PS, however vanishes with increasing wt% of ZnO. This may be due to the coordination bonding between aliphatic chain of PS and ZnO molecules. Due to the addition of ZnO into PS matrix, the intensity was increased and broadening of raman peaks was observed which depicts the coordination bond between filler and polymer. This peak was found to be similar as reported by Jeeju et al. 2011 [12].

Fig. 4 Raman spectra of PS-1 and PS-2 nanocomposite films

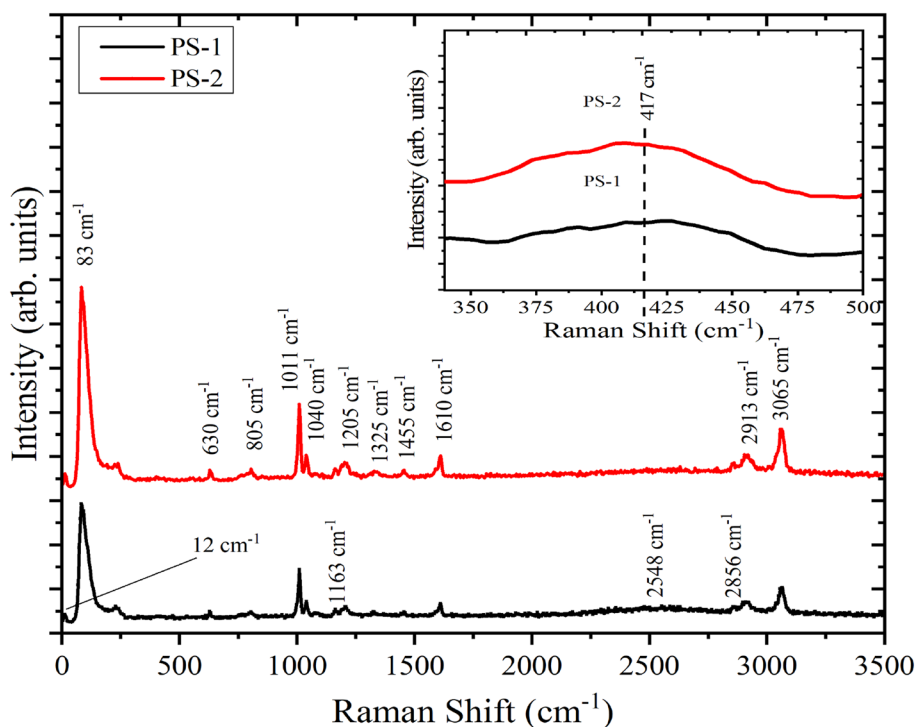
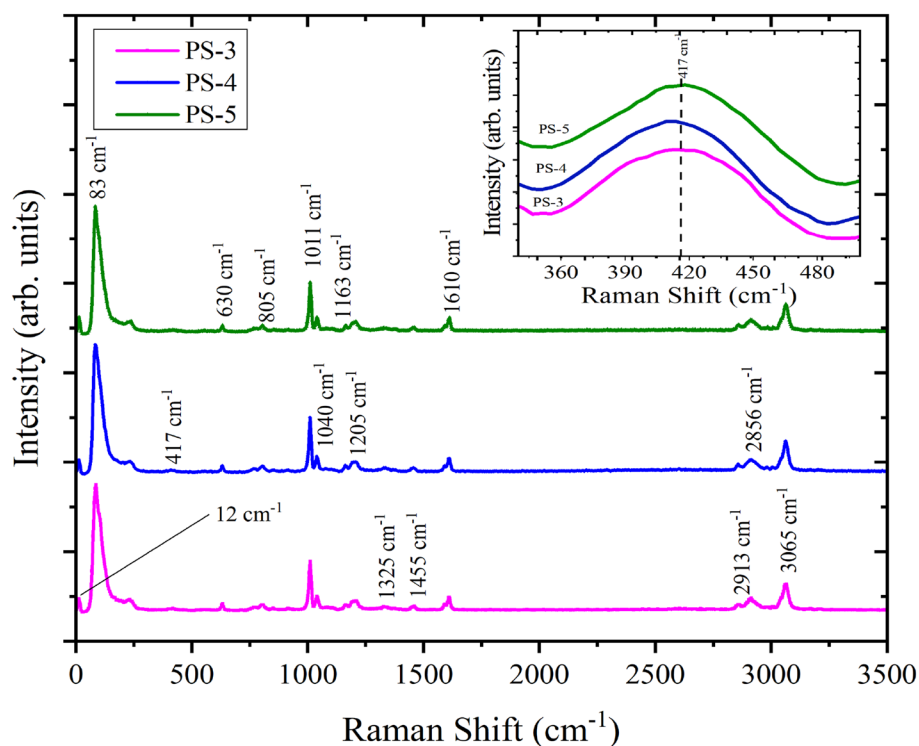


Fig. 5 Raman spectra of PS-3, PS-4 and PS-5 nanocomposite films



3.4 Photoluminescence (PL) Spectroscopy

The opto-electronic characteristics of the nanocomposite films can be studied via Photoluminescence spectroscopy. It is done by the emission of photoelectrons due to the interaction of incident light and the surface of the material. This emission phenomenon was recorded via photoluminescence spectrophotometer with an excitation wavelength of 350 nm at room temperature. Figure 6 shows the convoluted PL spectra of pristine and ZnO doped PS films. The separated shoulder and minor peaks for all the synthesized nanocomposite films are obtained by deconvolution of their PL spectra and depicted in Fig. 7a–e. For pristine PS films, strong emission peaks positioned at 384 nm & 403 nm in ultraviolet region and weak emission peaks positioned at 485 nm & 510 nm are observed. The emission in pure polystyrene film is due to the delocalization of electrons in the phenyl rings. Strong emission peak observed may be attributed to the recombination of electron hole pairs in intermediate trap states close to the valance band. The red shift in emission peak at 403 nm was observed at 404 nm, 405 nm, 406 nm and 410 nm by incorporating 5, 10, 15, and 20 wt% of ZnO into PS. This shifting may be due to the nucleating effect of ZnO or the presence of surface trap centers. The emission peak for pure PS observed at 384 nm disappears with the addition of 5 & 10 wt% ZnO into the polystyrene matrix but it again starts originating with the addition of substantial amount i.e. 15 & 20 wt% of ZnO. It might be due to the near band edge (NBE) emission of

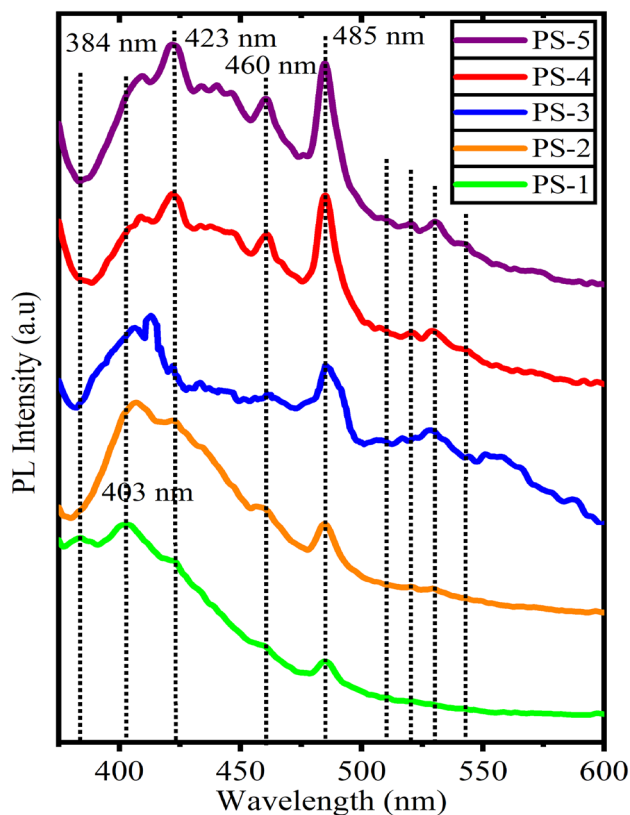


Fig. 6 Photoluminescence spectra of PS-1, PS-2, PS-3, PS-4 and PS-5 nanocomposite films

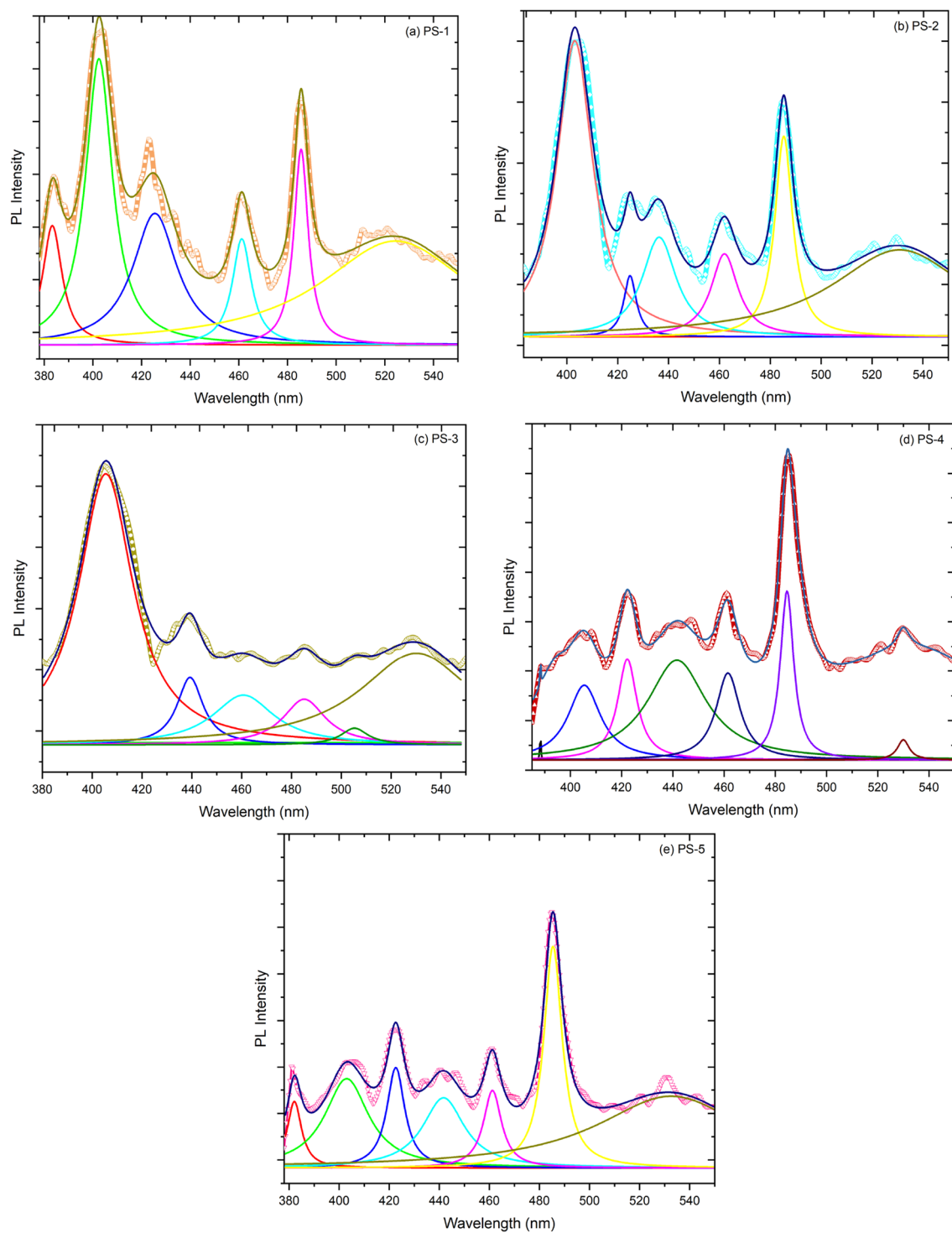


Fig. 7 a–e Deconvoluted photoluminescence spectrums of **a** undoped, **b** 5wt%, **c** 10wt%, **d** 15wt% and **e** 20wt% ZnO doped PS nanocomposite films

ZnO nano powder [26, 30]. The peaks located at 423 nm, 460 nm, and 485 nm starts dominating with the increased ZnO wt% due to the interfacial interaction of ZnO nanoparticles with PS matrix. However, in the wavelength region 510 nm -550 nm, the smoothness of the curve for pristine PS disappears with increasing ZnO concentration and few new peaks were observed at 510 nm, 521 nm, 531 nm, 544 nm. These small peaks and band edges observed in visible region may be considered as deep level peaks confirming the presence of structural defects such as oxygen vacancies or zinc and oxygen interstitials in nanocomposite films. This generation of defects might be attributed to the transition of electrons during emission from conduction band to various intermediate trap levels [31].

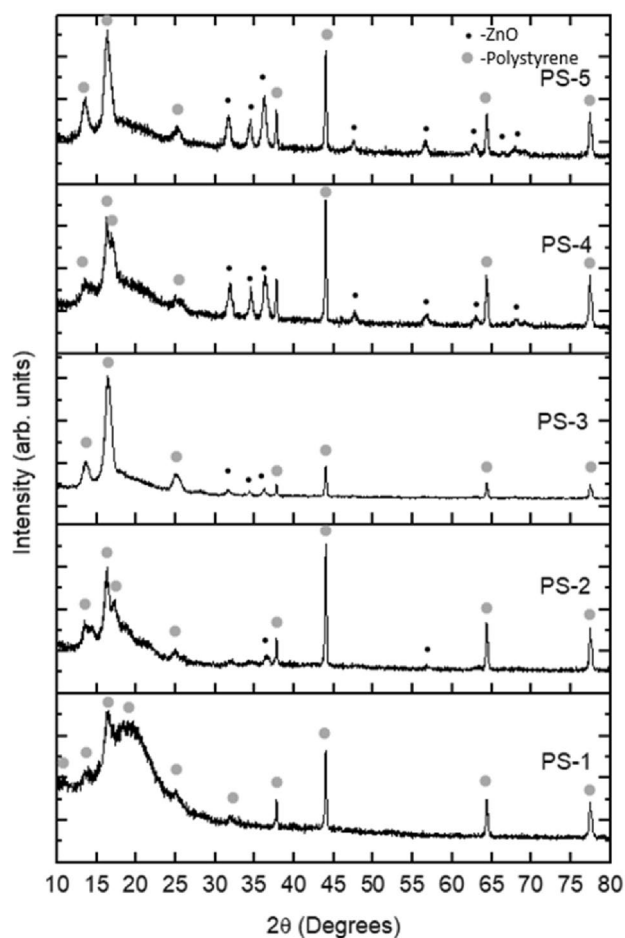


Fig. 8 XRD spectra of **a** PS-1, **b** PS-2, **c** PS-3, **d** PS-4 and **e** PS-5 nanocomposite films

3.5 X-Ray Diffraction Technique

X-Ray Diffraction technique gives useful insight to study the structure of polymeric materials. It helps us to provide information about crystallite size, degree of crystallinity, dislocation density and microstrain etc. present in the material. In the present study, XRD analysis of nanocomposite films was performed to investigate the effect of ZnO nano powder on the structure of PS polymer matrix.

Figure 8a shows the diffraction pattern of pristine PS. This spectrum has broad as well as sharp peaks. The broad peaks were positioned at (2θ values) 10.96° , 13.65° , 16.41° , 19.22° , 25.16° , 32.02° , which suggests the amorphous nature of PS film. Peaks positioned at 37.77° , 44.08° , 64.35° , 77.51° are sharp and confirms the presence of crystalline phase in pristine PS. For nanocomposite films, the XRD spectrums have peaks corresponding to both PS and ZnO confirmed from the JCPDS card Numbers 00-060-1507 (PD3) and 00-036-1451 respectively. Few new peaks were observed in the spectra of PS/ZnO nanocomposite films other than of pristine PS (Fig. 8b–e). A broad suppressed peak for pristine PS is appeared at 10.96° but disappear as we start the doping of ZnO into PS matrix. These broad peaks observed at 13.65° , 16.41° and 25.16° for pure PS becomes sharp & intense with the increase of ZnO from 5 to 20wt%. Peaks at 31.71° , 34.53° , 36.26° , 48° , 56.67° , 62.85° , 66° and 68.07° are shown in spectra of nanocomposite films. These peaks are attributed to the presence of ZnO into the PS polymer matrix corresponding to the miller indices (100), (002), (101), (102), (110), (103) and (112) respectively. They become dominant with the increase of ZnO wt%. These peaks were compared with the XRD peaks of ZnO nano powder and JCPDS file, which confirms the presence of ZnO. Peak at 62.85° which is invisible in pure PS & lightly doped PS i.e., 5% and 10% ZnO, becomes visible at 15% and 20% ZnO. A peak located at 32.02° is very weak in pure PS but becomes dominant as we increase the wt% of ZnO and peak intensity increases with increase in concentration of ZnO. The increase in peak intensity of nanocomposite films confirms more structural defects [7, 31]

The degree of crystallinity tells about the percentage of crystalline as well as long and short range order in the different materials like crystalline, semi-crystalline and amorphous. The increase in crystallinity of material results into the increase in hardness and density of the material. The degree of crystallinity can be calculated by different methods viz. peak fitting method, amorphous subtraction method etc. Here, the percentage of crystallinity among synthesized samples has been calculated using peak fitting method as

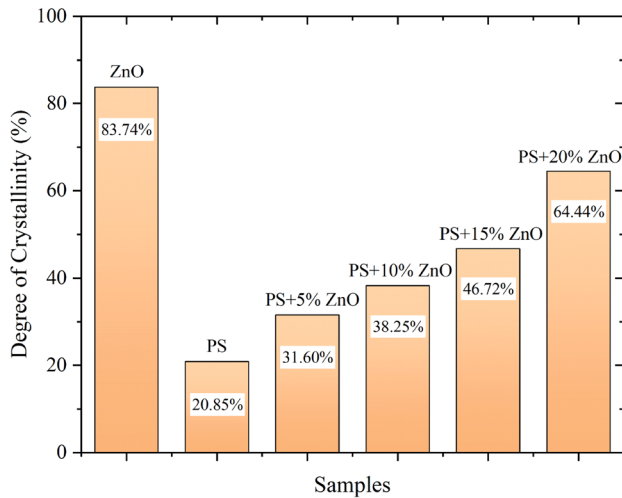


Fig. 9 Degree of crystallinity for synthesized nano powder and nano-composite film samples

$$\text{Crystallinity} = \frac{A_c}{A_c + A_a} \times 100$$

where A_c is the area of crystalline peaks and A_a is the area of amorphous peaks.

The calculated value of crystallinity for ZnO nano powder was 83.74% and for pristine PS was 20.85%. It shows the crystalline nature of ZnO and semicrystalline nature of pristine PS. It was confirmed from the presence of some sharp peaks in diffraction spectrum of PS matrix. The crystallinity of PS/ZnO nanocomposite films increases with the increase in concentration of ZnO among polymer matrix is depicted in Fig. 9. It leads to the increase in density of the film, which is in the good agreement with experimental and theoretical density measurements of the present study (Table 1).

XRD technique can also be used to calculate various parameters viz. full width at half maximum (FWHM), interplanar spacing (d) and crystallite size (D) corresponding to different 2θ values. These parameters can be calculated from Bragg's law of diffraction and Debye–Scherrer's formula [32] respectively as

$$2d\sin\theta = n\lambda$$

where θ is the diffraction angle, n is the order of diffraction and λ is the wavelength of X-rays.

$$D = \frac{0.9\lambda}{\beta\cos\theta}$$

where λ is the wavelength of X-ray (nm) and β is Full Width at Half Maximum (FWHM).

Dislocation density gives information about the defects present in a material. It is the ratio of length of dislocation lines to the volume of the crystal. The dislocation density (δ) and lattice strain (ϵ) can be calculated using crystallite size and FWHM as

$$\delta = 1/(D^2) \text{ and } \epsilon = \frac{\beta\cos\theta}{4}$$

The average values of interplanar spacing, crystallite size, dislocation density and micro strain were calculated for ZnO, pristine and nanocomposite films are shown in Table 3.

4 Conclusions

The PS/ZnO nanocomposite films were prepared successfully via solution mixing and casting process with different wt% of nano-sized ZnO. The opto-structural properties of the films were analyzed to make them applicable in optoelectronic and UV-shielding devices. The energy band gap of the nanocomposite films decreased from 4.45 to 3.01 eV with the increase in concentration of ZnO nano-filler. RAMAN and PL studies clearly depicts that the ZnO nano powder is completely gelled with PS solution forming nanocomposite films. The change in crystallinity from 20.85 to 64.44% is also observed using XRD and correlated to the density measurements. Change in band gap with change in ZnO concentration suggests the UV-shielding capability of 5wt% ZnO doped nanocomposite

Table 3 Calculated XRD parameters for ZnO nano powder, pristine and nanocomposite films

Samples	Average interplanar spacing (d)	Average crystallite size (D)	Dislocation density (δ) ($\times 10^{-3}$) (nm) $^{-2}$	Micro strain (ϵ) ($\times 10^{-3}$) (nm) $^{-2}$
ZnO nano powder	0.1876 nm	18.27 nm	3.51	2.01
PS-1	0.3326 nm	36.24 nm	43.23	4.31
PS-2	0.3106 nm	23.07 nm	8.58	2.55
PS-3	0.2826 nm	37.45 nm	4.27	1.73
PS-4	0.2474 nm	65.46 nm	4.48	1.79
PS-5	0.2552 nm	41.05 nm	3.49	1.75

film (PS-2). This nanocomposite film can be used as a best UV-shielding material because its transparency is very much comparable to pristine PS and has UV-absorber efficiency in the range of 280–320 nm.

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Declarations

Conflict of interest The authors declare that they have no known conflict financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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